Adaptive Thermal Fire Detection Node

Subject: This project focuses on the design and implementation of an adaptive IoT edge node for detecting possible fire-like thermal anomalies using a Raspberry Pi and an MLX90640 thermal camera. The system must collect thermal frames from the camera, process them locally, extract temperature-based features and publish the resulting measurements to an MQTT broker. The frame acquisition rate must be adaptive, allowing the node to operate at a low sampling rate under normal conditions and increase the sampling rate when suspicious thermal activity is observed. In case of a possible fire-like event, the system must publish an MQTT alert and, when necessary, transmit a thermal heatmap encoded in base64.

Objectives:

- Design an IoT edge monitoring system for thermal anomaly detection
- Integrate the MLX90640 thermal camera with Raspberry Pi using I2C
- Collect thermal frames periodically from the sensor
- Extract frame-level metrics such as minimum, average and maximum temperature
- Calculate hot pixel count, hot area ratio and temperature variation over time
- Implement adaptive sampling with normal, watch and alert operating modes
- Publish periodic telemetry messages through MQTT
- Generate alerts through MQTT when possible fire-like activity is detected
- Generate thermal heatmaps from the 32x24 thermal frames
- Transmit selected heatmaps through MQTT using base64 encoding
- Store measurements and alerts locally for evaluation
- Evaluate the system using controlled experiments and discuss limitations

Hardware list:

- 1x Raspberry Pi
- 1x MLX90640 Thermal Camera